Survival Rescue Procedures

3.8.2 (1) The manager of an underground mine shall establish a training program in survival mine rescue, including the use of approved self-rescue apparatus, and the use of fire fighting equipment in place at the manager's mine.

(2) All persons, employed at the mine or not, shall be instructed and trained in these procedures before entering the underground mine.

Instructor Emergency Procedures

3.8.3 The manager of a mine shall appoint a qualified person to instruct and establish a training program for all employees in basic emergency response, and fire fighting.

Fire Fighting

Fire Fighting Equipment

3.9.1 (1) The manager shall ensure that fire fighting equipment is provided and maintained at all locations at the mine where fire may endanger life.

(2) Unless specified otherwise in this code, The *British Columbia Fire Code* 1998 and subsequent supplements and revisions shall apply in determining the level of fire fighting equipment and maintenance as prescribed in subsection (1).

Underground Fire Fighting

Underground Fire Fighting

3.10.1 (1) The manager shall ensure that fire fighting equipment is provided and maintained at all underground crusher stations, electrical installations, pump stations, shaft stations, tipples, conveyors, service garages, fueling stations, and where a fire hazard may exist.

(2) A suitable number of fire extinguishers of a specified capacity shall be provided and maintained at each stationary diesel motor, transformer substation and any splitter panel.

(3) The location of the fire fighting equipment required by sections 3.10.1 (1) and (2) shall be such that, in the event of a fire, the direction of the mine ventilation air flow will not prevent or hamper the effective use of the equipment.

3.10.2 Fire extinguishers which are capable of giving off or generating poisonous gas when operated shall not be allowed below ground and in confined spaces.

3.10.3 The manager shall ensure that inspection and if necessary, testing and maintenance of all fire fighting equipment are carried out by a qualified person at least once a month.

Fire Fighting Plans

3.10.4 The position of all fire fighting pipe lines, hydrant valves, fire stations and fire cabinets shall be shown on a plan drawn to a scale of not less than 1:2500 with a metric scale bar and a marked north point. The plan shall be updated at intervals not exceeding three months.

Underground Coal Fire Fighting

Underground Coal Fire Fighting

3.11.1 For underground coal mines the following sections up to and including section 3.11.6 will apply in addition to all previous sections on fire fighting as may be applicable.

Minimum Requirements

3.11.2 A reservoir containing not less than 100,000 L of water at all times shall be maintained to supply water at adequate volume and pressure to underground coal mine workings.

Provision of Hydrants

3.11.3 Fire hydrants operated by wheel valves shall be located

(1) 20 m on the intake side of conveyor loading and transfer points, main junctions, and electrical substations,

(2) along such roadways as deemed necessary by the mine manager, at intervals not exceeding 100 m., and

(3) at suitable central points in room and pillar workings.

3.11.4 In close proximity to each fire hydrant there shall be a cabinet containing

(1) a branch pipe and nozzle of 40 mm minimum internal diameter, and

(2) sufficient lengths of hose to cover the distance between each cabinet, and the hose shall have a minimum internal diameter of 40 mm and a working pressure of 1000 kPa.

3.11.5 (1) A fire station shall be situated at the bottom level and at every intermediate level of a downcast shaft or slope which provides access to active working areas of the mine.

(2) In the case of a drift mine with a walkable intake airway, the manager may locate a fire station on surface close to that airway.

3.11.6 The minimum equipment housed in or near a fire station shall be

(1) not less than 120 m of fire hose with couplings, branch pipe, a nozzle of 40 mm minimum internal diameter designed to a working pressure of 1,000 kPa,

(2) a back-up supply of fire extinguishers, and

(3) a supply of sand bags or equivalent material.

Gas Detectors

Approved Detector

3.12.1 Every device used for the detection of flammable or noxious gas at a surface or underground mine shall be of a type approved for such use by a recognized testing agency.

Servicing Detectors

3.12.2 The manager shall appoint a qualified person to be responsible for maintaining the appliances used for detection of flammable or noxious gases.

Use of Appliance

3.12.3 The manager shall appoint a qualified person to carry and use an appliance for the detection of flammable or noxious gas. The qualified person shall

(1) check the appliance for damage and to ensure it is in proper working order before use,

(2) not use the appliance if found damaged or dysfunctional, and

(3) take all reasonable precautions to prevent the appliance from being damaged

Evacuation

Procedures and Instructions

3.13.1 The manager of a mine shall

(1) prepare procedures for the safe evacuation of personnel from the mine, or part of the mine, if necessitated by an emergency,

(2) post copies of the procedure in conspicuous places at the surface and underground, and

(3) ensure that each employee receives instruction in the procedures prepared under subsection (1) and that the employee can recognize the emergency warning system and is familiar with the emergency escape routes from the mine.

Emergency Warning System

3.13.2 The manager shall develop and maintain a system acceptable to an inspector for warning all employees, whether underground or in buildings on surface, of an emergency requiring prompt evacuation of their work places.

Test of System

3.13.3 A test of the warning system required under section 3.13.2 that does not involve evacuation of key process personnel shall be carried out at least once every 12 months on a production shift, and the manager shall ensure

that key process personnel unable to evacuate are knowledgeable with the warning system, and the evacuation procedure.

Report of Test

3.13.4 A report of all emergency warning system tests, including their effectiveness, shall be produced in a timely manner by the manager, reviewed by the OHSC and kept on file at the mine.

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Buildings – General

Design and Construction

4.1.1 (1) All buildings shall be constructed in accordance with the *British Columbia Building Code*.

(2) Where a building or structure is to be constructed, altered, dismantled, moved, or where major repairs are to be made, the manager shall ensure that

(a) drawings are prepared clearly showing all required field connections and any other information necessary for the safe completion of the work, or

(b) the work is supervised by a professional engineer.

(3) Drawings of the completed building or structure shall be kept at the mine and any revisions to the drawings shall be noted.

Proximity to Underground Workings

4.1.2 (1) Any building, whether or not attached to a headframe or other entrance to an underground mine, shall be of non-combustible construction for any portion that is within 15 m of the headframe or entrance to the mine.

(2) The remainder of the building shall be either of non-combustible construction or separated from the non-combustible section by a firewall, having a 4-hour fire resistance rating, and constructed in accordance with the *British Columbia Building Code*.

Portable Heaters

4.1.3 Every portable heater shall conform to the relevant sections of the following code, as amended from time to time

(1) CSA Standard B 139-00 "*Installation Code for Oil Burning Equipment*,"

(2) CSA Standard B 149.1-00 "N*atural Gas and Propane Installation Code*,"

(3) *Canadian Electrical Code*, and

(4) CSA Standard B51-97 "*Boiler, Pressure Vessel, and Pressure Piping Code*."

Access to Work Areas

4.1.4 (1) The manager shall ensure that all places where work is performed have safe means of access and egress and if necessary an alternate means of escape appropriate to the conditions of the work, and workers shall not use any other means that are, or may be, hazardous.

(2) Where required by an inspector, aisles and passageways in work and in storage areas shall be clearly delineated by floor markings. Such markings shall be maintained in a clearly visible condition

(3) (a) Emergency exits shall be clearly marked and designed to provide quick and unimpeded exit.
(b) doors shall not open directly onto stairways, but shall open onto floors or landings having a width in excess of the swing of the doors,
(c) double-acting swing doors shall be designed and installed to permit an adequate view through the doors, and
(d) transparent glass doors and glass panels shall be constructed of laminated, tempered, or wired safety glass meeting the current requirements of the *British Columbia Building Code*.

Equipment Entranceways to Buildings

4.1.5 (1) Any entranceway, other than for persons, through which equipment is moved shall
(a) exceed in width by at least 500 mm the width of the equipment where the width of the equipment is less than 3 m,
(b) exceed in width by at least 1 m the width of the equipment where the width of the equipment is 3 m or greater, and
(c) exceed in height by at least 300 mm the height of the equipment.

(2) All power-operated vertical doors installed in equipment entranceways shall have devices along the lower edge which will automatically stop movement of the door upon sensing an intermediate obstruction or when it reaches floor level.

Storage of Materials

4.1.6 (1) No material or equipment shall be so placed, stacked, or stored as to constitute a hazard to persons.

(2) No person shall enter a bin, stockpile area, or any place where there is a danger of becoming trapped in loose material unless safe access has been provided by catwalks, walkways, or other means, or unless the person is equipped with a safety belt and a secured lifeline and is attended by another person who is capable of effecting an immediate rescue.

(3) Any area upon which materials may be dropped or dumped shall be barricaded and posted with warning signs to prevent persons from inadvertently entering.

Handrails and Guardrails

4.1.7 (1) Every flight of stairs having more than 4 risers shall be equipped with handrails on all open sides of the stairways, and on one side of an enclosed stairway 1.12 m or less in width, and on both sides of an enclosed stairway over 1.12 m in width.

(2) The top of a handrail shall be at a height of 810 mm to 910 mm above the stair tread, measured vertically from the nose of the tread, and the height shall not vary on any flight or succession of flights of stairs.

(3) Handrails on open-sided stairways shall be fitted with midrails located approximately equidistant from the top of the handrail and the nose of the stair tread.

(4) When a stairway ends in direct proximity to dangerous traffic or other hazards, detour guardrails shall be installed.

Standard Guardrails

4.1.8 (1) Standard guardrails shall be installed where any open-sided floor, working platform, runway, walkway, or balcony is more than 1 m above grade or floor level.

(2) When persons are employed around open tanks, containing liquids or harmful substances, the sides of the tanks shall be constructed to extend at least 1 m above any working platform, or standard guardrails shall be provided to prevent persons from falling into the tanks.

Floor Openings

4.1.9 Openings or pits in floors, roofs, walkways or work areas accessible to persons, shall be securely covered or fitted with fixed, removable or collapsible guardrails.

Toe-boards

4.1.10 (1) Notwithstanding section 4.1.8, toe-boards are not required to be installed with a standard guardrail

(a) where persons do not or cannot work underneath, or

(b) where there is no machinery located below.

(2) Where materials are stored nearby, toe-boards shall be increased in height, or solid or mesh panels of appropriate height shall be installed to prevent any material from falling.

Walkways and Vehicle Curbs

4.1.11 (1) Walkways shall not be less than 500 mm in width and shall be provided with safe access by stairways or fixed ladders.

(2) Where necessary, curbs or bullrails not less than 250 mm in height, or similar means, shall be used to prevent vehicles or other equipment from running off the edge of an elevated surface.

Buildings – Surface

Repair Shops

4.2.1 (1) A repair shop shall be located such that site conditions favour adequate drainage away from the shop.

(2) Designated parking areas in the vicinity of the repair shop shall be clearly marked.

Coal Preparation Plant

4.2.2 Where combustible dusts may be continuously, or intermittently, in suspension in the air in coal preparation plants, a dust collecting system shall be provided and shall

(1) if constructed after this code comes into effect, be located away from the working area, outside the building or in a separate room properly ventilated to the outside,

(2) if a dry-type, be of non-combustible construction and equipped with adequate explosion doors or vents, and

(3) if a bag type, only have bags made of anti-static, fire resistant material and be provided with an effective electrical ground.

Dry-type Dust Collectors

4.2.3 A dry-type coal baghouse dust collector shall be provided with a device to

(1) detect the presence of hot or glowing particles upstream from the collector and trigger a quenching action, and

(2) sense a rise in temperature within the collector and sound an alarm and initiate an automatic quenching action at pre-set temperature levels.

Explosion Venting

4.2.4 (1) Provisions shall be made for venting an explosion from any coal preparation and dryer plant building that produces, or may produce, airborne dust in accordance with NFPA 120, Chapter 4 - “Fire and Explosion Prevention of Specific Hazards.”

(2) Ventilating hoods and exhaust ducts shall not be used as explosion venting devices unless they are properly designed for a dual purpose.

Monitoring of Explosive Atmospheres

4.2.5 Where explosive atmospheres may occur, such as in, above, or below a coal silo, continuous monitoring of the atmosphere shall be instituted.

Thermal Dryer Discharge

4.2.6 Thermal coal dryers that discharge to enclosed storage shall have equipment to continuously monitor coal discharge for hot or glowing embers of coal and, if detected, an automatic quenching system shall be activated.

Underground Mines – Enclosures, Service Areas, Fuel Stations, Pipelines and Storage

Underground Enclosures

4.3.1 All underground enclosures, including service garages, oil and grease storage areas and fuelling stations shall

(1) be designed and protected to prevent the inadvertent entry of an uncontrolled vehicle,

(2) have safe means of entry and exit appropriate for the conditions and purpose of the enclosure,

(3) be equipped with adequate fire protection, and

(4) be adequately ventilated to suit the purpose of the enclosure.

Underground Service Areas

4.3.2 Every underground service garage, fuelling station and oil and grease storage area shall

(1) be located so that a fire or explosion inside the facility would have a minimal effect on other working areas of the mine and on underground installations,

(2) be equipped with an automatically actuated fire suppression system that has been properly designed and installed to provide effective fire protection,

(3) be provided with means for manually actuating the fire suppression system from several locations inside the enclosure and at least one location outside the entrance, and whenever the fire suppression system is actuated, an alarm shall be given automatically at suitable, crewed locations on the surface of the mine and underground, or at a central fire control station,

(4) have a concrete floor without service pits,

(5) be equipped with means for containing spills of fuel, oil or grease including the use of fireproof receptacles into which the spilled fuel, oil or grease shall be deposited and removed from the mine without undue delay, and

(6) have adequate clearances for the safe performance of all work.

Underground Fuelling Stations

4.3.3 (1) The manager shall notify the inspector, in writing, of the intention to construct an underground fuelling station.

(2) No smoking shall be allowed in, or near, a fuelling station and signs to this effect shall be posted in conspicuous locations.

(3) A fuelling station shall

(a) be separate from service garage,

(b) incorporate a sill or curb to contain spilled fluids, and

(c) be totally enclosed, equipped with a self-closing door of non-flammable construction.

(4) Where a mobile fuelling supply tank is used, the tank shall be clearly labeled with "No Smoking" signs and no person shall smoke within 10 m.

Underground Oil and Grease Storage Areas

4.3.4 (1) The manager shall notify the inspector, in writing, of the intention to construct an underground oil and grease storage enclosure for quantities in excess of those outlined in section 4.3.10(2)(b).

(2) Every underground oil and grease storage enclosure shall be separate from a service garage.

(3) Oil shall not be stored in an underground mine in quantities greater than 500 L unless it is placed in an oil storage enclosure that

(a) meets the requirements of sections 4.3.1, 4.3.2 and 4.3.3 of the code,

(b) is provided with a door,

(c) has a two-hour fire resistance rating, and

(d) is maintained in proper order and kept free from obstruction so as to be readily usable at all times.

(4) No service garage or fuelling station shall be installed underground unless it meets the appropriate requirements of sections 4.3.1, 4.3.2 and 4.3.3 of the code, and the manager has established a safe operating procedure.

Fuel Pipelines Underground Construction

4.3.5 Where a fuel pipeline is used in an underground mine it shall

(1) be constructed of minimum or better standard weight wrought iron or steel pipe or the equivalent with respect to strength, durability, corrosion and fire resistance,

(2) have leak proof joints, and if sealants or gaskets are used, they shall be approved by Underwriters Laboratories for fuel oil service,

(3) be designed, installed, and used in accordance with the manufacturer's specifications, and

(4) be drained empty after completion of each fuel transfer.

Installation

4.3.6 A fuel pipeline in an underground mine shall

(1) be installed to minimize the risk of damage and supported so as to avoid dips and sags,

(2) be clearly identified as a fuel pipeline,

(3) be pressure tested before initial use to 345 kPa above atmospheric, or 1.5 times the maximum working pressure, whichever is greater,

and the pipeline shall retain the pressure for at least 2 hours after the source of pressure has been removed,

(4) be visually inspected at least once a month, and

(5) not pass through garages, switch rooms, explosive magazines, or refuge stations.

Fuel Storage Tanks Construction

4.3.7 Fuel storage tanks in a fuel transfer pipeline shall

(1) be constructed of steel and designed in accordance with good engineering Practice for their location and use, and NFPA 122 "Storage of Flammable and Combustible Liquids Within Underground Metal and Non-Metal Mines (Other than Coal) 1986" may be used for the design of the tanks except where any requirement conflicts with a requirement of this part of the code,

(2) be supported and anchored to prevent excessive stress concentration on any supporting portion of the shell, and located to minimize the risk of damage, and

(3) have vent pipes of sufficient size, located so that fumes are directed away from any place where they could be a hazard to health or safety.

Fire Protection and Control

4.3.8 Every fuel storage tank shall

(1) have a means of accurately determining the amount of fuel it contains,

(2) be identified as to its contents and that it is a fire hazard area, and

(3) be surrounded by a dike or curb capable of containing 110% of its storage capacity.

Fuel Oil Transfer Systems

4.3.9 (1) A fuel oil transfer system shall have electrical equipment complying with Section 20 of the *Canadian Electrical Code*, Part 1.

(2) It shall be equipped with a fire suppression system and have adequate fire extinguishers.

(3) It shall be designed and installed so that

(a) only a pre-set quantity of fuel can be transferred at one time and this quantity shall be less than 90% of the available storage capacity of the receiving tank at the time of the transfer, or

(b) when the receiving tank is full, a sensing device will stop the flow of fuel at the sending tank, or

(c) one person at the sending tank and one at the receiving tank shall have suitable communication so that the flow of fuel can be stopped at any time, and the controls and switches at the tanks shall be clearly marked.

(4) Where fuel transfer lines are installed in a shaft, fuel transfer shall not be conducted during hoisting operations.

(5) An authorized person shall be responsible for ensuring that the system is regularly inspected, cleaned and properly maintained, and that all procedures are strictly followed.

General Requirements

4.3.10 (1) Oil, grease, liquid fuels, and other flammable materials shall not be kept or stored within 30 m of a shafthouse, mine portal or other mine opening, and the natural drainage from the storage area shall be away from the shaft or mine entrance.

(2) Oil, grease, and flammable liquids used in an underground mine with a flashpoint below 52 degrees Celsius shall

(a) be transported and stored only in metal containers or receptacles or in portable plastic containers approved for petroleum fuels, and

(b) when stored underground, unless in an approved enclosure, be restricted to a sufficient quantity for

(i) the current day's work in the case of volatile, flammable liquids, and

(ii) 7 days in the case of oil and grease.

4.3.11 (1) No device for the generation of acetylene gas and no internal combustion engine that uses gasoline, propane or any other volatile substance as a fuel shall be used in an underground mine.

(2) Except when used for burning or cutting, propane or other similar fuel that is heavier than air when in a gaseous state shall not be allowed underground.

(3) When propane or other similar fuel that is heavier than air is being used underground for burning or cutting, the containers for the fuel shall be of a type certified under the *Energy Efficiency Act* and shall not have a capacity greater than 10 kg.

4.3.12 Flammable materials shall only be stored in areas approved by the manager and designed for such storage.

Machinery and Equipment with Moving Parts

4.4.1 [Renumbered as 4.5.01]

Guards

4.4.2 (1) In this section and sections 4.4.2.1 to 4.4.2.8, 4.4.16, 4.5.3 and 4.11.6, **"guard"** means a physical barrier that provides protection from a hazard.

(2) The manager must ensure that machinery or equipment with moving parts is equipped with guards which are designed, constructed, installed and maintained so that

(a) no person can reach over, under, around or through the guard and access a hazard including a drive belt, chain, rope or pulley, sprocket, flywheel, geared wheel, any opening through which a belt, pulley or wheel operates, any bolt, key or set screw, every part of any wheel or other moving part that projects unevenly from the surface, or all head, tail, drive and tension pulleys of a conveyor, and

(b) all material that may be ejected in the normal course of operating the machinery or equipment, which could be hazardous to a person, is safely contained.

(3) Subsection (2) of this section, and sections 4.4.2.1 to 4.4.2.8, do not apply to idlers of a conveyor belt except with respect to return idlers, beneath a conveyor belt, that

(a) are directly above a regular walkway or work area, and

(b) present a risk of harm to a person on the walkway or in the work area, beneath the return idlers, by being vertically reachable or by detaching and falling.

Fixed Guards

4.4.2.1 (1) In this section, **"fixed guard"** means a guard that is kept in place with either a permanent attachment system or an attachment system that necessitates the use of tools to remove or open the guard.

(2) The manager must ensure that, when fixed guards are removed from machinery or equipment, the attachment systems of the fixed guards remain attached to the guards or to the machinery or equipment.

Interlocked Guards

4.4.2.2 (1) In this section, **"interlocked guard"** means a guard that is attached, and interlocked, to the operational controls of machinery or equipment which prevents the operation of hazardous machinery or equipment functions.

(2) The manager must ensure that interlocked guards comply with CSA Z432:23, Chapter 9.5, Guards, interlocked.

Distance Guards

4.4.2.3 (1) In this section, **"distance guard"** means a guard that prevents access to a hazard by its physical dimensions and distance from the hazard.

(2) The manager must ensure that distance guards

(a) have no greater than a 0.15 m opening between the adjacent walked surface and the bottom of the distance guard,

(b) have a minimum height of at least 1.8 m,

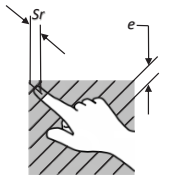
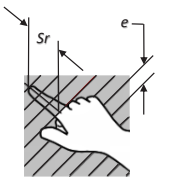
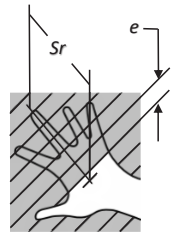
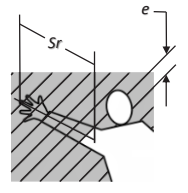
(c) open laterally or away from the hazard and cannot close by itself, and
(d) can only be removed by use of a tool.

Construction of Guards

4.4.2.4 The manager must ensure that all guards
(a) are designed to be of a suitable size and weight to permit ease of handling,
(b) are of robust construction and are free of sharp edges, rough surfaces and protruding parts that could cause injury,
(c) cause a minimal amount of visual obstruction during the production process, and
(d) meet the requirements of Table 4-1 for opening size and distance from a hazard.

Table 4-1

Reaching through regular openings (in mm) – Persons 14 years of age and above*

Part of Body	Illustration	Opening	Safety Distance to Hazard zone, S_r		
			Slot	Square	Round
Fingertip		$e \leq 4$	≥ 2	≥ 2	≥ 2
		$4 < e \leq 6$	≥ 10	≥ 5	≥ 5
Finger up to knuckle joint		$6 < e \leq 8$	≥ 20	≥ 15	≥ 5
		$8 < e \leq 10$	≥ 80	≥ 25	≥ 20
Hand		$10 < e \leq 12$	≥ 100	≥ 80	≥ 80
		$12 < e \leq 20$	≥ 120	≥ 120	≥ 120
		$20 < e \leq 30$	≥ 850	≥ 120	≥ 120
Arm up to junction with shoulder		$30 < e \leq 40$	≥ 850	≥ 200	≥ 120
		$40 < e \leq 120$	≥ 850	≥ 850	≥ 850

*This table is based on data from Table 9.1 of CSA Z432:23

Servicing Machinery or Equipment in Motion Without Removing Guards

4.4.2.5 Machinery or equipment with guards may be serviced while in motion, without the lockout procedures set out in sections 4.11.1 to 4.11.7 applying, if the manager ensures that

(a) the guards remain in place, and

(b) only the parts of the machinery or equipment that are necessary for servicing are energized.

Testing or Adjusting Energized Machinery or Equipment

4.4.2.6 If machinery or equipment with guards can only be tested and adjusted after the guards are removed and while the machinery or equipment is energized, the lockout procedures set out in sections 4.11.1 to 4.11.7 do not apply to the machinery or equipment while it is being tested or adjusted provided the manager

(a) establishes a safe procedure for testing and adjusting the machinery or equipment in accordance with the manufacturer's recommendations, if any,

(b) ensures that only qualified persons, or employees under the direct supervision of a qualified person, perform the testing and adjusting, and

(c) ensures that the safe procedure referred to in paragraph (a) is reviewed by the qualified persons and employees at the frequency determined by the manager.

Machinery or Equipment Isolated by Height

4.4.2.7 Despite section 4.4.2, guards are not required for exposed moving parts of machinery or equipment if the exposed moving parts are 3 m or more above the ground, working surface, walkway, stockpiles or material build-up, whichever surface is closest to exposed moving parts.

Overhead Drive Belts

4.4.2.8 Despite section 4.4.2.7, the manager must ensure that an overhead drive belt has guards if the whipping action caused by the belt breaking may be hazardous to a person.

Grinders

4.4.3 (1) A grinder shall be assembled, adjusted and operated in accordance with the manufacturer's specifications.

(2) The maximum speed at which a grinding wheel may be operated shall be indicated on the grinding wheel or be easily obtainable.

(3) A grinding wheel shall be

(a) enclosed by a protective hood except for the area at the workrest,

(b) stored where it will not be damaged by impact, extreme heat or cold, and

(c) stopped when the grinder or workrest is being adjusted.

(4) The operator of a grinder shall wear a face shield complying with the relevant requirements of CSA Standard Z94.3.99.

(5) The workrest on a grinder shall be mounted above the centre line of the grinding wheel and not more than 3 mm from the wheel unless otherwise specified by the manufacturer.

(6) An air-operated grinder shall have a governor to prevent it operating in excess of the rated speed of the grinding wheel.

(7) The governor required by subsection (6) shall be inspected regularly and maintained in proper condition.

(8) Powered grinding wheels, other than portable hand held machines, shall be equipped with an exhaust system or other means for removing dust produced during the grinding operation.

Chain Saws

4.4.4 Chain saws shall meet the requirements of CSA Z62.1-95 "Chain Saws".

Pneumatic Tools

4.4.5 (1) Portable pneumatic nailing and stapling tools capable of driving a fastener having a nominal diameter exceeding 1.2 mm (18 gauge ASWG), shall be designed so that the operator is required to make not less than two separate operations to activate the tool. One of these operations shall be to place the tool against the work surface.

(2) Under no circumstance shall the operating trigger be taped or otherwise secured in the "on" position, or held in the "on" position while moving between operations.

(3) The air supply shall be disconnected before the tool is serviced or any adjustments made.

(4) The safe operating air pressure specified by the manufacturer for tools, hoses and fittings shall not be exceeded.

4.4.6 Repealed.

Welding and Burning Equipment

4.4.7 (1) Whenever torches or welding equipment are being transported or used, the manager shall require that safe procedures are developed and followed.

(2) Every compressed gas cylinder containing fuel gas or other gas, when not in use, shall

(a) be stored in an upright position and kept in a well ventilated area free from flammable material and electrical apparatus, and

(b) have a securely fastened, approved valve protection cap, and be adequately secured by chains or other acceptable means to prevent it from being knocked over.

(3) At any time that compressed gas cylinders are being transported, they shall be secured on a special carrier or container designed for the purpose, and unless the carrier or container is of a type intended for safely moving gas cylinders, the cylinders shall be fitted with securely fastened, approved valve protection caps.

Powder-actuated Tools

4.4.8 A powder-actuated fastening system, consisting of the tool, power loads and fasteners shall meet the requirements of ANSI Standard A10.3-1995, "American National Standard for Construction and Demolition Operations - Safety Requirements for Powder-Actuated Fastening Systems"

Lifting Devices

4.4.9 (1) A lifting device shall be designed, constructed and installed in accordance with recognized standards and good engineering practice, provided with overwind protection if power operated, and provided with an identification plate.

(2) The maximum load that a lifting device may carry shall be established by its designer.

(3) A notice showing the maximum established load shall be posted in a location visible to the operator and, except during testing, it shall not be exceeded.

(4) Each component that may affect the safe operation of a lifting device shall be examined and tested by a qualified person before initial use and thereafter at intervals not exceeding one year, and a record shall be kept showing the dates, findings and names of the qualified persons performing the examinations and tests and the record shall be kept available for inspection.

(5) Where a combination of lifting devices is used simultaneously, the work shall be supervised by a qualified person

Install Fire Protection

4.4.10 (1) Fire extinguishing equipment of suitable type and size shall be provided at all fire hazard areas.

(2) A fire suppression system consisting of sprinklers, foam or other suitable means of suppressing fire shall be provided

(a) in an underground mine on fixed equipment containing more than 10 L of flammable fluids actuation of the fire suppression system shall also cause power shut off, and

(b) on the surface, in a building or structure, except a fan house, located above or adjacent to an opening to an underground mine.

Boilers and Engines Near Underground Workings

4.4.11 No part of a steam boiler or stationary internal combustion engine shall be installed within 30 m of any part of the collar of a shaft or openings to underground workings.

Engine Exhausts

4.4.12 (1) The exhaust of an internal combustion engine which is temporarily or permanently installed within a building on the surface shall be conducted to a point outside the building and prevented from re-entering the building, entering the intake of any compressor, contaminating the atmosphere of another building and contaminating mine workings.

(2) No internal combustion engine shall be used in potentially hazardous areas, as defined in the *Canadian Electrical Code*, Part 1, of surface mine buildings, unless the manager has obtained permission, in writing, from an inspector.

Fans for Supplying Underground Ventilation

4.4.13 All fans supplying ventilating air to underground workings and all structures containing fans shall be of non-combustible construction, installed at the surface except in the case of auxiliary or development fans, and

(1) provided with means for reversing the direction of air flow, and

(2) the flow shall not be reversed without the manager's authorization.

Main Surface Ventilating Fans

4.4.14 (1) Where explosive atmospheres may exist in underground workings, the main surface ventilating fans shall

(a) be offset not less than 5 m from the nearest side of the mine opening and be equipped with noncombustible air ducts,

(b) be provided with explosion doors, and

(c) be operated from a dedicated power circuit.

(2) Notwithstanding subsection (1), a fan may be located directly in front of, or over, a mine opening if

(a) the opening is not in a direct line with possible air blasts coming out of the mine in the event of an explosion, and

(b) there is another opening not less than 5 m, or more than 30 m, from the fan opening that is in a direct line with possible air blasts and is equipped with explosion doors.

Elevators

4.4.15 (1) All elevators shall be installed and maintained in accordance with CSA B44-00 "*Safety Code for Elevators*".

(2) All aerial lifts shall be installed and maintained in accordance with CSA B31 1-M1999 "*Safety Code for Manlifts*."

(3) Prior to initial operation, and thereafter at 12-month intervals, all drive components shall be subjected to non-destructive tests.

(4) A maintenance record shall be kept listing all unsafe conditions reported and the repairs made to each elevator to correct unsafe conditions. This record shall be retained at the mine until abandonment or removal of the elevator.

Conveyor Belts

4.4.16 (1) No person shall ride on a conveyor belt.

(2) No person shall cross a conveyor belt except at an established foot bridge not less than 500 mm in width equipped with guardrails.

(3) Every conveyor way shall be provided with a walkway or other acceptable access for maintenance and inspection purposes.

(4) Every accessible section of a conveyor shall be provided with a pull cord to stop the conveyor in an emergency and the controls shall be arranged so that they have to be reset manually before the conveyor can be restarted after an emergency stop.

(5) On every conveyor which can be started automatically by remote control or where the operator has limited visibility of the whole conveyor, an audible start up warning device shall be installed and there shall be a time delay of at least 10 seconds between the end of a minimum 10 second warning and conveyor start up.

(6) Repealed.

(7) A belt conveyor used underground, or a belt conveyor more than 15 m in length installed in a building, or other closed-in structure, shall be provided with a belt slip detection device to stop the drive motor in the event of belt blockage or slippage, and when required by the inspector, with an effective sprinkler system and plugged chute switches which shall stop the conveyor when a plugged chute condition occurs.

Cleaning Up Spillage

(8) Servicing, or cleaning up spillage, on or around a moving conveyor belt shall only be carried out

(a) where the conveyor system is so constructed that the work can be done safely and, subject to subsection (8.1), without removing any guards, and

(b) by persons who have been fully trained and authorized by the manager to do the work.

(8.1) If a conveyor system has guards that are not required under section 4.4.2, subsection (8) of this section applies to servicing, or

cleaning up spillage, on or around the moving conveyor belt as set out, except that the guards may be removed.

(9) When it is necessary to remove guards for servicing or cleanup, the conveyor shall be stopped and locked out in accordance with sections 4.11.1 to 4.11.7 of this code.

(9.1) Subsection (9) of this section only applies with respect to guards that are required under section 4.4.2.

(10) All guards removed during cleanup or servicing shall be replaced before the locks are removed and the conveyor is started.

(11) The manager shall develop safe work procedures for any work near moving conveyors and submit any major or significant changes to established safe work procedures to the chief inspector for approval and these procedures shall

(a) address specific problems associated with each conveyor at the mine and indicate the speed at which each conveyor travels, and

(b) for cleanup of spillage, include a safe procedure or mechanism for return of material to a moving belt, and a procedure or mechanism to allow the removal of materials lying below the conveyor that protects persons from contact with the moving parts of the conveyor and any material that may fall from the conveyor.

(12) Conveyor belting for use in the transportation of coal or in an explosive atmosphere and in all underground locations shall meet the requirements of CSA Standard CAN/CSA-M422-M87 "Fire Performance and Antistatic Requirements for Conveyor Belting" or an equivalent standard subject to approval by the chief inspector.

Steam and Compressor Plant Operation

4.4.17 (1) The manager shall ensure that procedures for the safe operation of a steam or compressor plant are prepared in writing and made available to persons operating and maintaining the plant.

(2) Steam and compressor plants shall conform to the requirements of the *Safety Standards Act*, Power Engineers, Boiler, Pressure Vessel and Refrigeration Safety Regulation.

(3) A steam boiler or compressor to which the *Safety Standards Act*, Power Engineers, Boiler, Pressure Vessel and Refrigeration Safety Regulation does not apply shall be maintained in a proper and safe condition by a qualified person.

Pressure vessels

4.4.17.1 Pressure vessels and associated components as defined in CSA B51-14 must be installed, certified, repaired and maintained in conformance with that standard.

Compressors Carbon Monoxide Monitor

4.4.18 (1) An air compressor driven by a prime mover exceeding 30 kw, that is lubricated by oil and discharges to a closed system at a pressure greater than 100 kPa, shall have temperature shutdown devices installed at the high-pressure discharge pipe.

(2) All compressed air systems that are used to supply underground workings shall incorporate a carbon monoxide monitoring system that continuously monitors the compressed air located between the high-pressure discharge and the receiver.

(3) The carbon monoxide monitoring system shall, when the carbon monoxide sampled reaches 25 ppm, sound an alarm and cause the immediate shutdown of all the compressors connected to the air supply system being sampled.

(4) The manager shall ensure that the carbon monoxide monitoring system is working by testing the system at least monthly and recording the results in the compressor maintenance record.

Underground Mine Heaters

4.4.19 (1) The installation and maintenance of mine air heating equipment authorized by section 6.35.1 of the code, including provisions for mounting, clearances, and air supply, shall conform to the requirements of the following standards as amended from time to time

(a) CSA Standard B 139-00, "*Installation Code for Oil Burning Equipment*,"

(b) CSA Standard B 149.1-00 "*Natural Gas and Propane Installation Code*."

(2) With the exception of embedded pipes or ducts, all parts of the heating system shall be readily accessible for inspection, maintenance, repair, and cleaning.

(3) A carbon monoxide detector, capable of detecting concentrations below 25 ppm, shall be installed to shut down the heater if carbon monoxide in excess of 25 ppm is detected. This detector shall be installed 15 m downstream from where the heated air enters the mine.

(4) Pipelines with gas pressures in excess of 3.45 kPa shall not be located within 15 m of any mine opening.

(5) Pressure regulating stations shall be clearly marked and protected from physical damage.

(6) Propane storage tanks shall be located such that a leak of its contents shall not enter any mine openings.

(7) A vibration switch shall be mounted on the fan motor that will shut down the heater at prescribed vibration limits.

Portable Ladders

4.4.20 Portable ladders shall meet the requirements of CSA Standard CAN3-Z1 1-M81 “Portable Ladders” or other equivalent standard.

Construction of Portable Ladders

4.4.21 Where portable wooden ladders are constructed and used at a mine

(1) the side rails shall measure not less than 38×89 mm (nominal 2"×4") for ladders up to 5 m in length and not less than 38×140 mm (nominal 2"×6") for ladders over 5 m in length,

(2) the cleats (rungs) shall be solid lumber not less than 21×89 mm (nominal 1"×4") spaced not more than 300 mm apart, secured in place by nails and with filler blocks of equal thickness to the cleat,

(3) the width between the side rails shall not be less than 380 mm or more than 500 mm,

(4) a double cleat ladder shall have 3 side rails, evenly spaced, be not less than 1070 mm and not more than 1270 mm in total width, and have cleats that extend the full width of the ladder, and

(5) material used for a ladder shall be of #2 grade or better from the following groups: Douglas Fir-Larch, Hem-Fir, Spruce-Pine-Fir or Coast-Sitka-Spruce; all lumber shall be graded to the National Lumber Grades Authority Rules or other approved grading rules.

4.4.22 Only transparent, protective coatings shall be applied to wooden ladders.

4.4.23 A portable single or extension ladder shall be held, tied or otherwise secured against slipping. The ladder shall extend at least 1 m above any upper landing to which it provides access.

Fixed Ladders

4.4.24 Fixed ladders shall meet the requirements of the American National Standards Institute (ANSI) Standard A14-3-1992 “Safety Requirements for Fixed Ladders” or other equivalent standards.

Piping Systems Buried Lines

4.4.25 (1) Piping systems containing substances hazardous to workers because they are corrosive, toxic, flammable or containing an expansible fluid, as defined in the *Safety Standards Act*, Power Engineers, Boiler, Pressure Vessel and Refrigeration Safety Regulation, under pressure shall be identified by a labeling or marking system.

(2) Workers shall be trained to identify the labeling.

(3) The marking or labeling shall be maintained in a legible condition.

4.4.26 Any new installations and where identifiable, any existing buried pipelines, power cables and storage tanks shall be plotted on accurate